

Biswarup Karmakar

Department of Computational and Data Sciences

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EDUCATION

- **Indian Institute of Science** Bengaluru, India
M.Tech. (Research) + Ph.D. in Computational and Data Science August 2022 – May 2026
Supervisor: Dr. Ratikanta Behera
Thesis Title: Tensor Generalized Inverses and Low-Rank Representation with Applications
Coursework CGPA: 8.7/10
- **Indian Institute of Technology Kanpur** Uttar Pradesh, India
M.Sc. in Mathematics August 2018 – June 2020
Supervisor: Dr. Prosenjit Roy
Project Title: Sobolev Spaces and Existence of Weak Solutions for Second-order Elliptic PDE
- **Serampore College, University of Calcutta** West Bengal, India
B.Sc. in Mathematics July 2015 – June 2018

RESEARCH INTERESTS

Tensor Decompositions, Tensor Generalized Inverses, Numerical Optimization, Machine Learning.

RESEARCH PUBLICATIONS AND PREPRINTS

1. **B. Karmakar** and R. Behera, *Efficient iterative methods for computing generalized inverse of tensors based on t -product*, Computational and Applied Mathematics, Vol. 44 (2025), No. 7, Paper No. 380, 29 pp. doi:10.1007/s40314-025-03339-z.
2. K. Panigrahy, **B. Karmakar**, J. K. Sahoo, R. Behera, and R. N. Mohapatra, *Computation of M -QDR decomposition of tensors and applications*, Journal of Applied Mathematics and Computing, Vol. 71 (2025), S437–S462. doi:10.1007/s12190-025-02487-z.
3. **B. Karmakar** and R. Behera, *Robust low-rank tensor completion based on M -product with weighted correlated total variation and sparse regularization*, minor revision submitted in “Applied Mathematics and Computation”. arXiv:2604.13525.
4. **B. Karmakar**, N. Bhadala, and R. Behera, *A family of iterative methods for computing generalized inverses of quaternion matrices and its applications*, under review. arXiv:2605.01464.
5. **B. Karmakar** and R. Behera, *Nonlinear transform-induced nuclear norm for quaternion tensor completion*, under review. arXiv:2605.01467.

ACHIEVEMENTS AND FELLOWSHIPS

- Qualified Graduate Aptitude Test in Engineering (GATE) Mathematics 2021 with AIR (All India Rank) - 58 (Score 683).
- Qualified Joint CSIR-UGC Junior Research Fellowship (JRF) with AIR - 125, June 2019 & AIR - 165, December 2019. (not availed)

- Qualified Joint Admission Test for M.Sc. in IITs (JAM) with AIR-194, 2018.
- Awarded INSPIRE Scholarship (top 1% students in Class XII Board Exam across India), Department of Science and Technology (DST), Government of India, 2015–2020.

TECHNICAL SKILLS

- **Languages:** Python, MATLAB, C, C++.
- **Libraries:** Pytorch, Tensorly, CUDA (cuBLAS), scikit-learn.
- **Software/OS:** Linux, Windows, LaTeX, MS Office.

RELEVANT COURSEWORK

Numerical Linear Algebra, Numerical Optimization, Introduction to Computing for AI and Machine Learning, Tensor Computations for Data Science, Random Variates in Computation, Introduction to Scalable Systems, Numerical Solutions of Differential Equations.

TEACHING EXPERIENCE

- Teaching Assistant, *Tensor Computations for Data Science* (M.Tech. CDS), Indian Institute of Science, Bengaluru, January – April 2024.
- Teaching Assistant, *Tensor Computations for Data Science* (M.Tech. Sponsored Programme, Online), Indian Institute of Science, Bengaluru, January – April 2024.
- Teaching Assistant, *Numerical Methods* (M.Tech. CDS), Indian Institute of Science, Bengaluru, August – November 2023.

REFERENCES

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Dr. Predrag S. Stanimirović

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Dr. Jajati Keshari Sahoo

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